CANARA ENGINEERING COLLEGE
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
VISEMESTER B.E. DEGREE I-IA SCHEME OF EVALUATION, SEPTEMBER 2019

COMPUTER ORGANIZATION MAX 50 MARKS

Q. No	Composition of Solution or Key Solution	Marks
1.a	DIAGRAM: Bus directions	1
	Registers	1
	Memory	1
	Control unit and ALU	1
	Explanation: Use of MAR, MDR, PC, IR, Rn	1+1+1+1+1
1.b	Diagram	1
	Explanation	1
	Two Advantages	1+1
	Two disadvantages	1+1
1.c	Definition of clock cycle	1
	Definition of clock rate	1
	Terms in performance equation N, S and R	1+1+1
1.d	About SPEC	1
	SPEC standards	1
	Meaning of SPEC rating	1
	SPEC rating formula for single program	1
	SPEC rating formula for multiple programs	1
1.e	Little Endian assignment definition	1
	Little Endian assignment diagram	1
	Little Endian assignment example	1
	Big Endian assignment definition	1
	Big Endian assignment diagram	1
	Big Endian assignment example	1
1.f	Byte addressability definition	1
	Byte addressability diagram	1
	World alignment definition	1
	World alignment diagram	1
2.a	Negative flag	1
	Zero flag	1
	Overflow flag	1
	Carry flag	1
2.b	Explanation and Example of Immediate addressing mode	1+1
	Explanation and Example of Register addressing mode	1+1
	Explanation and Example of Direct addressing mode	1+1
	Explanation and Example of Indirect addressing mode	1+1
	Explanation and Example of Index addressing mode	1+1

Q. No	Composition of Solution or Key Solution	Marks
2.c	Diagram showing Stack region and Stack Pointer	1
	Explanation of PUSH	1
	Explanation of POP	1
	Assembly routine for safe pushes (Compare SP, Branch for full error, Push)	1+1+1
	Assembly routine for safe pop (Compare SP, Branch for empty error, Pop)	1+1+1
2.d	Rotate right without carry (Explanation, Diagram and Example)	1+1+1
	Rotate left without carry (Explanation, Diagram, and Example)	1+1+1
	Rotate right with carry (Explanation, Diagram, and Example)	1+1+1
	Rotate left with carry (Explanation, Diagram, and Example)	1+1+1
3.a	Diagram (Bus lines, address decoder, control circuit, data & status register)	1+1+1+1
	Function of Address decoder	1
	Function of Control circuit	1
	Function of Data & status register	1
3.b	Use of DATAIN, DATAOUT, STATUS and CONTROL	1+1+1+1
	Program: Read a character	1
	Store the character	1
	Displaying it	1
	Repeating if no carriage return	1
4.a	Definition of are interrupt	1
	Definition of Interrupt Service Routines	1
	Definition of are interrupt latency	1
4.b	Raising an interrupt	1
	Processor getting interrupted	1
	Disabling interrupt request INTR	1
	Interrupt acknowledge INTA	1
	Executing ISR	1
	Enabling interrupt Request INTR	1
4.c	Daisy chain: Explanation, Diagram, advantage	1+1+1
	Priority groups: Explanation, Diagram, advantage	1+1+1

Signature of Course Instructor with date | **Signature of course coordinator/Expert with date** | **Head of the Department with date**